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SUPPLEMENTARY MATERIAL

The BBP Formula and the Coupling Polynomial on One Ladder

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Abstract
The coupling polynomial $f(K,s) = 1 + sK + K^2$ of Gravity Field Dynamics takes the values $\Phi_6(K), \Phi_4(K), \Phi_3(K)$ at $s = -1, 0, +1$, so its three states form one row of the ladder $X \to X^2$ of cyclotomic polynomials. We show that the Bailey–Borwein–Plouffe (BBP) formula for π lies on the same ladder, in its Field Origin column. In the variable $u = y/\sqrt{2}$ the denominator of its integral is $16(1-u)(1+u)(1+u^2)(1+u^4)$, three rungs Φ_2, Φ_4, Φ_8, which are the Field Origin polynomial at three consecutive levels and take the values $5, 17, 257$ at $K = 4$. The rungs are the binary coordinates of the cube, the top one being the half-turn. Every formula for π of BBP's type must drop the rung at the coupling row's own level, Φ_4, together with one half of Φ_8, because their poles carry the primes 3 and 5; π is read from the other half of $\Phi_8 = \Phi_4(X^2)$, the Field Origin one level up. Silence on the TIME slot and closure of the weights both act on the one remaining factor, and both select BBP.

Keywords: BBP formula, cyclotomic polynomials, coupling polynomial, Field Origin, stellated octahedron, proof

Setting

Let D be the positive integer with $2(D+1) = 2^D$, so $D = 3$ and $K = D + 1 = 4$ [6]. The *coupling polynomial* is $f(K,s) = 1 + sK + K^2$, read at $s = +1$ (TetA), $s = 0$ (the Field Origin) and $s = -1$ (TetB). Φ_n is the n-th cyclotomic polynomial, the minimal polynomial of $e^{2\pi i/n}$.

The *slot series* are $S_j = \sum_{k\ge 0} 16^{-k}/(8k+j)$, $j = 1, \dots, 8$. A *formula of type* $(*)$ for a number c is a rational vector w with $\sum_j w_j S_j = c$. Expanding $1/(16 - y^8)$ on $[0,1]$ gives

$$\sum_j w_j S_j = \int_0^1 \frac{p_w(y)}{16 - y^8}\, dy, \tag{1}$$

with $p_w(y) = 16 \sum_{j=1}^{8} w_j\, y^{j-1}$. BBP is $w_{\mathrm{BBP}} = (4, 0, 0, -2, -1, -1, 0, 0)$ [1]. With $a = (8, -8, -4, -8, -2, -2, 1, 0)$, every formula of type $(*)$ for π is $w(r) = w_{\mathrm{BBP}} + r\, a$, $r \in \mathbb{Q}$ [2, 4]. The proof of that statement [4] writes each slot as

$$S_j = -\tfrac{1}{8} \sum_{\omega^8 = 1} (\sqrt{2}\,\omega)^j\, L_\omega, \tag{2}$$

with $L_\omega = \mathrm{Log}\left(1 - \bar{\omega}/\sqrt{2}\right)$, and shows, with Baker's theorem [3], that the seven logarithms $i\pi$, $\log 2$, $\log(1+\sqrt{2})$, $\log(2 \pm i)$, $\log(\sqrt{2} \pm i)$ in which every L_ω is expanded are linearly independent over the algebraic numbers. We use these two facts.

The Coupling Polynomial Is a Row of the Ladder

Theorem 1. *$f(X,+1) = \Phi_3(X)$, $f(X,0) = \Phi_4(X)$ and $f(X,-1) = \Phi_6(X)$. Under $X \to X^2$,*

$$\begin{aligned} &\Phi_3(X^2) = \Phi_3(X)\,\Phi_6(X), \\ &\Phi_4(X^2) = \Phi_8(X), \qquad \Phi_6(X^2) = \Phi_{12}(X). \end{aligned} \tag{3}$$

So the rows (Φ_6, Φ_4, Φ_3) at $X = K^{2^n}$ are $(13, 17, 21)$, $(241, 257, 273)$, …, and the Field Origin column $\Phi_4(K^{2^n}) = 1 + K^{2^{n+1}}$ is $17, 257, 65537$.

Proof. The roots of $X^2 \pm X + 1$ and $X^2 + 1$ are the primitive roots of unity of orders 3, 6 and 4. The square roots of a primitive n-th root are primitive $2n$-th roots when n is even, and one primitive n-th and one primitive $2n$-th root when n is odd. □

The cyclotomic form of f and this ladder are already in the archive [5]; Theorem 1 states them as the frame for what follows.

BBP Lies on the Field Origin Column

Lemma. *For base 2^{D+1} and period 2^D, the poles of $1/(2^{D+1} - y^{2^D})$ lie at radius $2^{(D+1)/2^D}$. The radius is $\sqrt{2}$, and the poles include $1 + i$, exactly when $2(D+1) = 2^D$.*

Proof. $(D+1)/2^D = 1/2$ is the stated equation, and $(1+i)^8 = 16$. □

Theorem 2. *With $y = \sqrt{2}\,u$,*

$$16 - y^8 = 16\,(1-u)(1+u)(1+u^2)(1+u^4), \tag{4}$$

and the rungs $1 + u^{2^k}$, $k = 0, 1, 2$, are $\Phi_2(u), \Phi_4(u), \Phi_8(u)$, the Field Origin polynomial at three consecutive levels: $\Phi_2(X) = \Phi_4(\sqrt{X})$, $\Phi_4(X)$ and $\Phi_8(X) = \Phi_4(X^2)$. At $X = K$ they are $5, 17, 257$. Neither Φ_3 nor Φ_6 divides the denominator: of the coupling row, only Φ_4 occurs.

Proof. $1-u^8=(1-u)\prod_{k=0}^{2}(1+u^{2^k})$ by repeated difference of squares, and $1+u^{2^k}$ is $\Phi_{2^{k+1}}$. The factors of $1-u^8$ are the Φ_d with $d \mid 8$, and $3, 6 \nmid 8$. □

Theorem 3. *The rungs are the binary coordinates of the cube.* $\prod_{k=0}^{2}(1+u^{2^k})=\sum_{m=0}^{7}u^m$, *each* $m=4b_2+2b_1+b_0$ *arising once, so* $m\mapsto(b_2,b_1,b_0)$ *is a bijection onto the corners* $\{0,1\}^3$. *On the poles* $y=\sqrt{2}\,\omega$, *with* $\omega=e^{i\pi/4}$,

$$\omega^m=(\omega^4)^{b_2}(\omega^2)^{b_1}\omega^{b_0}, \qquad \omega^4=-1, \quad \omega^2=i, \tag{5}$$

so the three bits are the half-turn, the quarter-turn and the eighth-turn, and flipping the most significant bit multiplies every pole character by -1.

Proof. Each $m<8$ has one binary expansion; expand the product and the power. □

In (2) slot j carries the character ω^j, so Theorem 3 is the place-value reading of slot j as corner j mod 8 used in [4]. The ladder fixes which bit is which rung. It does not fix which spatial axis carries which bit, and the selection of BBP does not depend on that choice; whether the slot is counted as j or $j-1$ does matter [4].

Every Formula for π Drops the Coupling Row's Rung

Theorem 4. *If* w *is a formula of type* $(*)$ *for* π, *then* $(y^2+2)(y^2+2y+2)$ *divides* p_w. *Here* $y^2+2=2\Phi_4(u)$, *the Field Origin rung at the coupling row's level, and* $y^2+2y+2=2(u^2+\sqrt{2}u+1)$, *the half of* $\Phi_8=(u^2-\sqrt{2}u+1)(u^2+\sqrt{2}u+1)$ *with poles at* 135° *and* 225°. *The numerators are exactly*

$$\begin{aligned} p_{w(r)} &= 16\,(y^2+2)(y^2+2y+2) \\ &\quad\times\big[r\,(y^2-4y+2)-(y-1)\big]. \end{aligned} \tag{6}$$

Proof. By (2), $\sum_j w_j S_j=-\frac{1}{8}\sum_\omega P_w(\sqrt{2}\omega)\,L_\omega$ with $P_w(z)=\sum_j w_j z^j=z\,p_w(z)/16$. Write $\omega_n=e^{i\pi n/4}$. Of the seven logarithms, $\log(\sqrt{2}+i)$ occurs only in L_{ω_2}, $\log(\sqrt{2}-i)$ only in L_{ω_6}, $\log(2-i)$ only in L_{ω_3} and $\log(2+i)$ only in L_{ω_5}, each with coefficient 1 [4]. The sum equals $\pi=-i\cdot i\pi$, which has no component on these four logarithms, and the coefficients $-\frac{1}{8}P_w(\sqrt{2}\omega_n)$ are algebraic. By the independence over the algebraic numbers, each is zero. So p_w vanishes at $\sqrt{2}\omega_n=\pm i\sqrt{2}$ and $-1\pm i$, the roots of y^2+2 and y^2+2y+2. Dividing the numerators of $w_{\mathrm{BBP}}+r\,a$ gives (6). □

The reason is arithmetic. Over each conjugate pair of poles, the product of $1-\bar{\omega}/\sqrt{2}$ is $\frac{1}{2}$ at 0° and 180°, $\frac{1}{2}$ at 45° and 315°, $\frac{3}{2}$ on the Φ_4 pair and $\frac{5}{2}$ on the 135° pair. Only the last two contain an odd prime, $3=(\sqrt{2}+i)(\sqrt{2}-i)$ and $5=(2+i)(2-i)$, and logarithms of those primes cannot combine into π. The cancellation is forced on the whole family; it is not particular to BBP.

π Is Read One Rung Up

Theorem 5. *After the cancellation in Theorem 4 the integrand of every formula for* π *is*

$$\frac{16\big[r\,(y^2-4y+2)-(y-1)\big]}{(y^2-2)(y^2-2y+2)}, \tag{7}$$

whose poles are $\pm\sqrt{2}$, *on* $\Phi_1\Phi_2$, *and* $1\pm i$, *on the* 45° *half of* Φ_8. *Only the poles* $1\pm i$ *contribute to* π. *For BBP the* $1\pm i$ *part integrates to* $\pi+\ln 4$ *and the* $\pm\sqrt{2}$ *part to* $-\ln 4$. *The rung that carries* π *is* $\Phi_8=\Phi_4(X^2)$, *the Field Origin one level above the coupling row.*

Proof. The integrand follows from (6). At $\pm\sqrt{2}$, $1-\bar{\omega}/\sqrt{2}=1\mp 1/\sqrt{2}$ is real, so L_ω is real. At $1\pm i$, $1-\bar{\omega}/\sqrt{2}=(1\pm i)/2$, whose logarithm is $-\frac{1}{2}\log 2\pm i\pi/4$. The two integrals are elementary. □

Corollary. *The TIME slot* w_7 *is the coefficient of* y^6 *in (6),* $16r$; *the sum of the weights is* $p_{w(r)}(1)/16=-15r$. *So silence on the TIME slot and closure, weights summing to zero, are each the condition* $r=0$, *which leaves the factor* $-(y-1)$ *and is BBP. The factor* $y-1$ *is the closure condition.*

Scope

The results are exact. The coupling row (Φ_6,Φ_4,Φ_3) at K and BBP's Field Origin column (Φ_2,Φ_4,Φ_8) meet only in Φ_4. Every formula of type $(*)$ for π drops that rung, and π is read from the rung above it.

The type $(*)$, base $16=K^2$ and period $8=2^D$, is an input taken from [4]; the Lemma shows that it is the type whose poles reach $1+i$. Nothing here says that BBP is the stellated octahedron. No symmetry of the stellated octahedron has order 8, so BBP's period is not one of its symmetries; the ladder places that period one rung up. The primes 3 and 5 are not given a meaning in Gravity Field Dynamics. The rungs at $X=K$ multiply to $5\cdot 17\cdot 257=\sum_{m<8}K^m$, while $f(K,+1)=\sum_{m<3}K^m$; that is the geometric series, recorded only as an observation.

References

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